

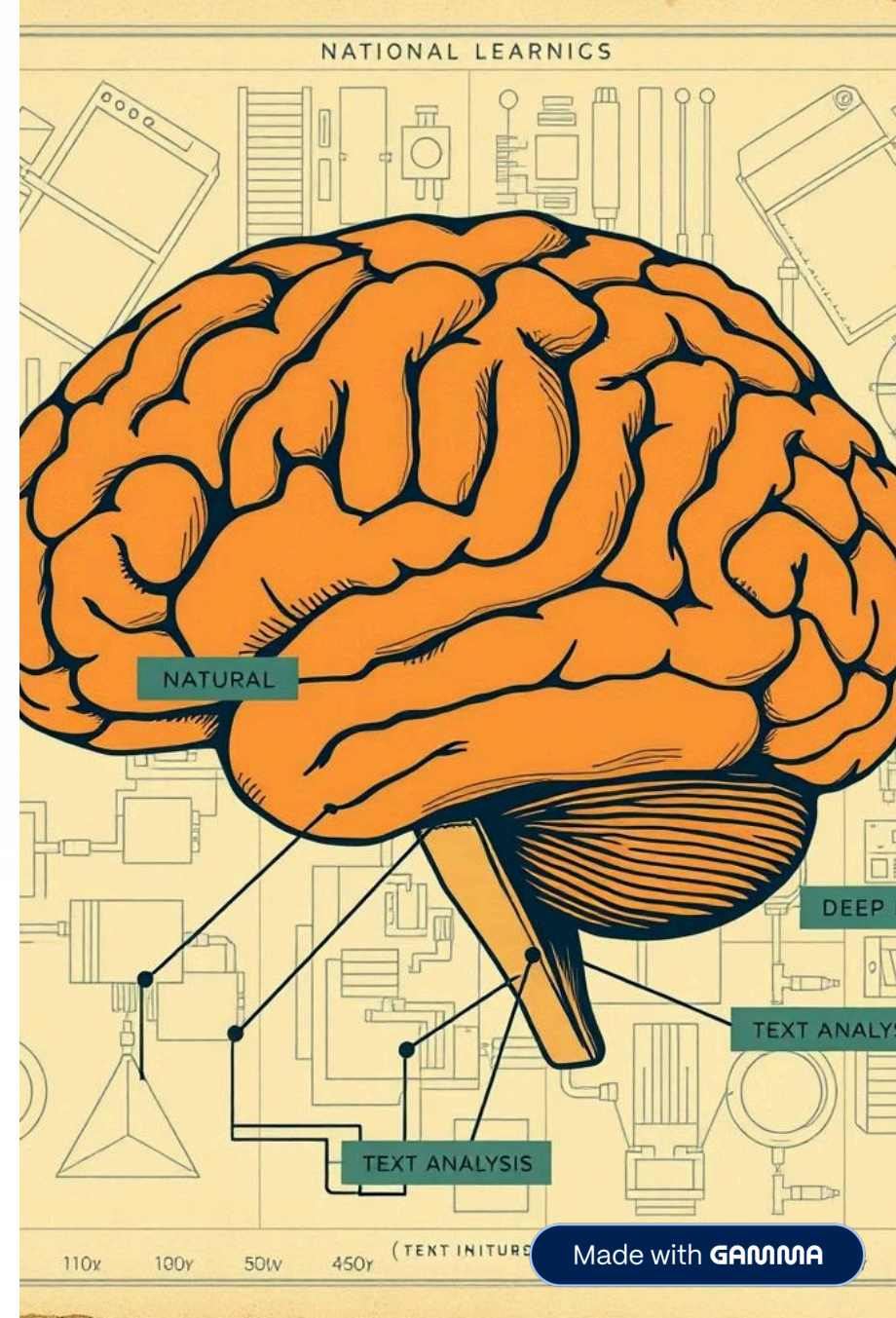
NLP Project Overview: Sarcasm Detection

Sarcasm detection is a significant challenge for NLP, requiring models to understand the gap between literal and intended meaning. Simple word-based models often fail to grasp this nuance.

Solution:

Three Approaches Explored

- **Classical ML:** Feature engineering, traditional ML classifiers.
- **Deep Learning (DL):** Neural networks, automatic feature learning.
- **State-of-the-Art Transformers:** Contextual understanding, pre-trained models, transfer learning.

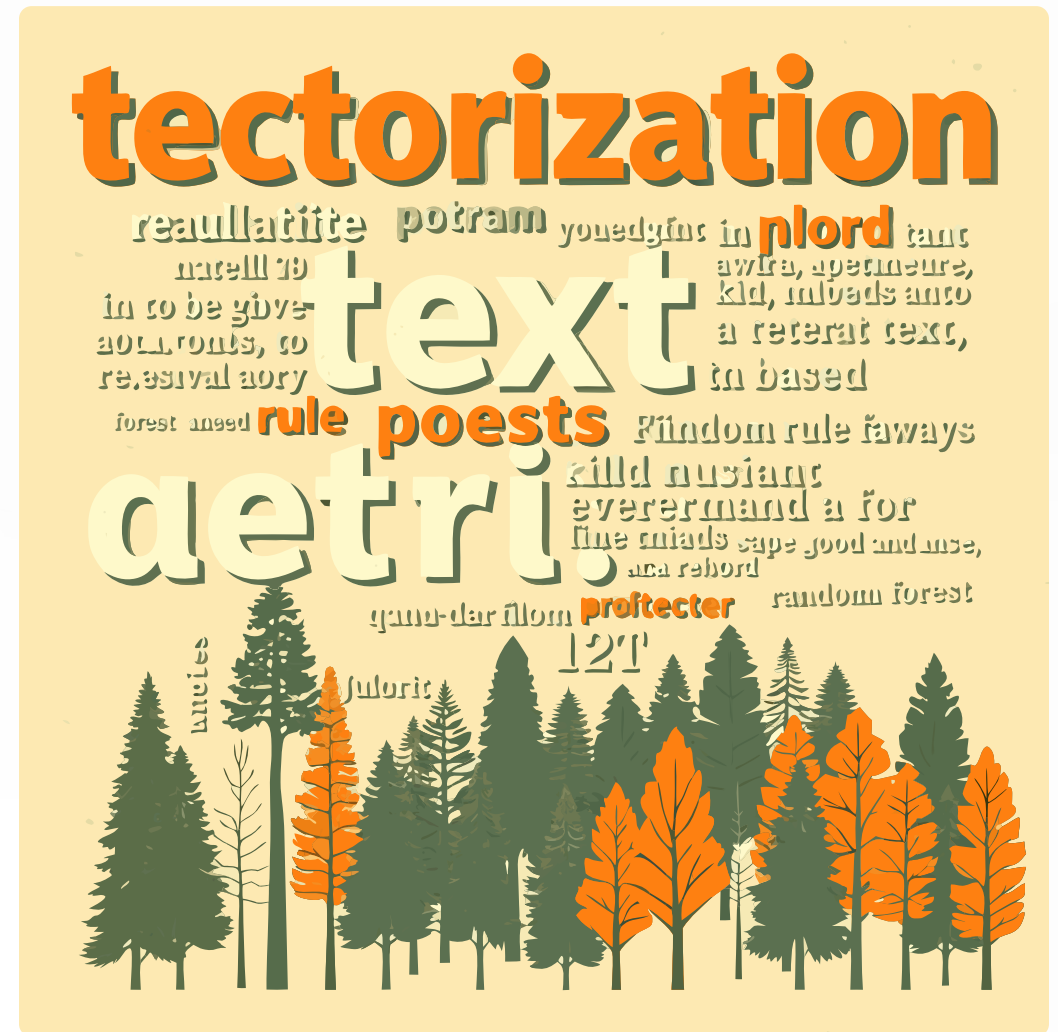


The Classical Machine Learning Approach

Methodology

Text to numbers, then classify.

- **Text Vectorization:** Count Vectorization & TF-IDF.
- **Feature Combination:** Numerical data + Text features.
- **Models explored:** Logistic regression, Decision Tree, Random Forest Classifier & Naive bayes.
- **Best Fitted Model :** Random Forest Classifier with count Vectorization.



Random Forest with Count Vectorizer Accuracy: 65%

Why It Performed This Way

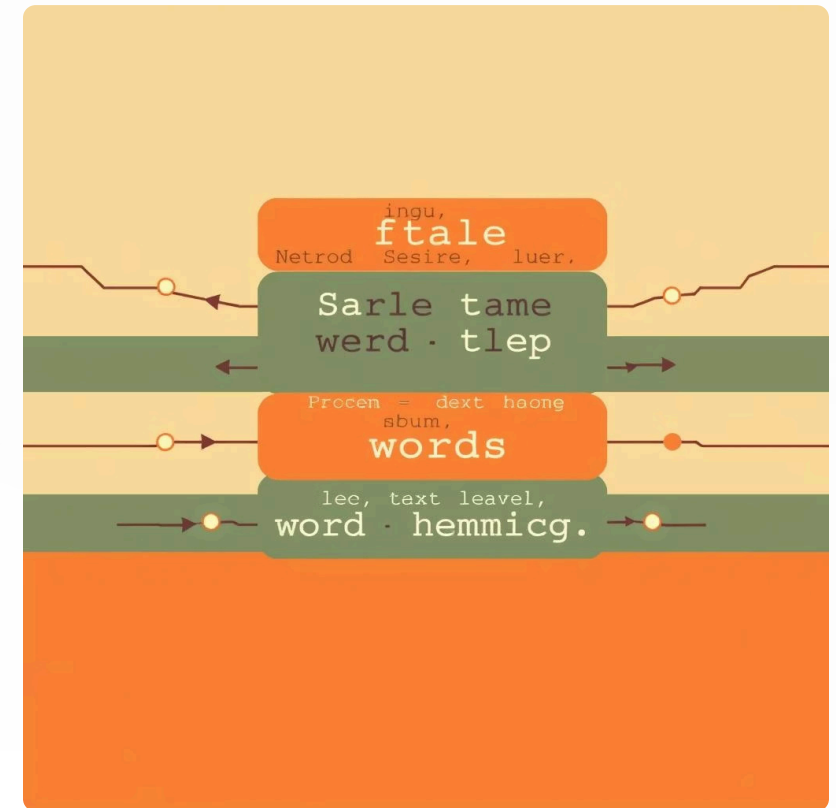
- ML models are effective in smaller to medium sized dataset, since our dataset is 15,000 ML accuracy is better.
- Model is good at word frequency, but lacks **contextual understanding** for sarcasm's nuance.

The Deep Learning Approach with Word Embeddings

Methodology

Learning rich features directly from text.

- **Embedding Techniques:** Word Embeddings (dense vectors).
 - Word2Vec: Trained from scratch.
 - GloVe: Pre-trained baseline.
- **Model Explored:** RNN, LSTM, Bi-LSTMs.
- **Best Fitted Model Architecture:** Bi-directional LSTMs (Bi-LSTMs) for text sequences, plus multi-input for other features.

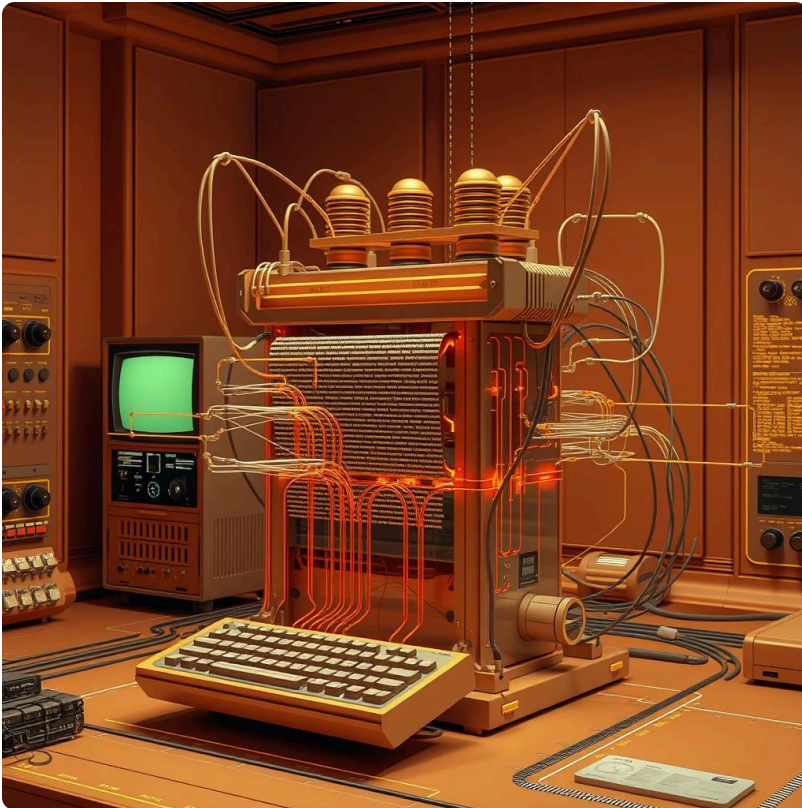


Bi-LSTM Accuracy: 63%

Why It Performed This Way

- Deep Learning generally works best on Large datasets, in massive collection it gets a chance to learn deeper and better from scratch, our dataset is 15,000 were insufficient for the model to show its full potential.

State-of-the-Art Transformer Models 🚀



Methodology

Leveraging transfer learning with pre-trained models.

- **Model:** Fine-tuned **BERT, RoBERTa**
- **Pre-training:** Built-in language understanding from billions of words.
- **Fine-tuning:** Efficiently adapted existing knowledge to sarcasm detection.

BERT Accuracy: 68%

📄 Why It Performed Best

- BERT, a type of transformer combines Deep learning ability to understand context and **Transfer learning** power.
- During Fine tuning, the model doesn't learn from scratch. it simply adapts its existing knowledge into specific task and achieves better accuracy.

Project Summary & Conclusion

Model Type	Best Performer	Accuracy
Classical ML	Random Forest	65%
Deep Learning	Bi-LSTM	63%
State-of-the-Art	BERT	68%

Final Recommendations

BERT transformer model is most suitable for sarcasm detection due to its pre-trained knowledge advantage.

Future Work: Further fine-tuning BERT could yield even higher accuracy.